Patrik **Drbal**

Master's Student in Theoretical Computer Science · Software Developer

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Education

M.S. in Theoretical Computer Science — *Prague*

Faculty of Information Technology CTU in Prague — *Jun. 2023 - Jun. 2026*

- So far, my master's studies have been accompanied by a close cooperation with the Department of Theoretical Computer Science and their activities. I also participate actively in several student clubs. For more information, please see the following sections.
- For my Master's thesis, I am working on TODO.

Exchange studies — *Finland*

Aalto University — *Jan. 2025 - Jun. 2025*

- During my exchange studies, I regularly attended events held by the local TCS community, namely weekly CS Theory Seminars and monthly Frontier Fridays.

B.S. in Computer Science — *Prague*

Faculty of Information Technology CTU in Prague — *Sep. 2019 - Jun. 2023*

- In my Bachelor's thesis, Hiding Leaders in Covert Networks: A Computational Complexity Perspective, I explored hardness of the Hiding Leaders graph problem under the parameterized computational complexity framework.

Work Experience

Graduate Teaching Assistant — *Prague*

Dept. of Theoretical Computer Science, FIT CTU in Prague — *Sep. 2024 - present*

- Leading exercise sessions of an introductory course to graph theory, algorithms and data structures.
  - Autumn 2024 - 2 weekly sessions
  - Autumn 2025 - 1 weekly session

Software Developer & Analyst — *Prague*

Target Technologie — *Jul. 2022 - present*

- My tasks on this position include:
  - Designing and integrating new functionalities to the in-house 2D nesting (two-dimensional (irregular) bin packing) library. As 2D bin packing is a geometric, NP-complete, optimization problem, my work involves dealing with geometric and algorithmic, efficiency-oriented tasks, for which we use JTS Topology Suite and JGraphT graph library, together with various discrete math techniques.
  - Server management – security, backuping, maintatence.
  - Solo-developing a new 2D geometry visualization library for the company's purposes.
  - Other tasks like refactoring, testing or code reviews.

Warehouse auxiliary worker — *Netherlands*

OTTO Work Force — *Jul. 2018 - Aug. 2018*

- When I was 18, I went to Netherlands for my summer job. I worked as an auxiliary worker in several warehouses across Netherlands for five weeks.

Leisure time educator - basketball — *Sušice, Czech Republic*

Children and Youth Centre — *Oct. 2017 - Jun. 2019*

- For two school years, me and my friend lead a leisure time basketball club for kids between the ages of 11 and 18.

Extracurricular Activity

Study and Research Activities

- **GGOAT Research Group.** I participate actively in events organized by the GGOAT (Graphs, Games, Optimization, Algorithms, Theoretical Computer Science) research group.
  - **STIGMA.** I participated in STIGMA (School on Theoretical Informatics, Graphs And Mathematical Applications) in years 2024 and 2025, where I gave a ninety-minute-long talk in english about edge deletion to tree-like graph classes and solution discovery via reconfiguration, respectively.
  - **PACE Challenge.** I participated in PACE (Parameterized Algorithms and Computational Experiments) 2024 Challenge about the One-Sided Crossing Minimization problem. Although I was not a core member of our team, I contributed by reading through some of the literature and commiting to our codebase.
  - **Monday Seminars.** I regularly visit monday seminars organized by the group, which are similar to the Aalto ones.
- **ATHENS Programme.** In Autumn 2023, I took part in the ATHENS Programme in Madrid, where I visited a week-long course called Big Data and Geo-Data Science: Python Programming for Geospatial Processing, Analysis, and Query.

Student Clubs and Associations

- I am an active member of various student clubs and associations. I believe it's an important part of life at university.
  - **FIT++ (Faculty Student Club).** I co-organized a Linux workshop for freshmen students in Autumn 2023. I occasionally help preparing club's events such as board games nights or open study sessions.
  - **Sincoolka (Dormitory Student Club).** I am one of the dormitory's network administrators and a maintainer of the club's clubroom. I regularly attend board meetings and other club gatherings.
  - **CHEESE Sitsit.** In Autumn 2025, I was one of the core organizers of an international event in Prague. We hosted about 80 people for one weekend with the main event was a sitsit.
- At high school, I participated in the following activities:
  - **Programming, Robot and Linux Clubs.** I attended two after-school clubs; Game Maker Game Engine Programming, and Lego Mindstorms Robot Construction and Programming. With the latter, we also took part in the FIRST LEGO League competition a couple of times, where disciplines Robot Programming and Design, Research Project, and Team Cooperation were involved. In my last year, I was also teaching Linux basics to a couple of interested students.
  - **Technical support.** For four years, I was helping the student theatre club as a lighting and sound technician. I also provided technical support during other school events.
  - **Basketball Community.** I was a core part of the local basketball community. With my friend, we actively encouraged young people in our home city to play basketball by teaching them and by organizing open basketball sessions, where anyone could come and play. We were also connecting new and younger players with larger community of older and even professional players, by doing which, we provided them enough information, contacts and opportunities to play basketball on a more serious level. Our efforts lead to a semi-professional basketball career of one of the former members of the community, while many others continue to play basketball recreationally to these days.